

FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2022

TEST DATE: 09-01-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A paramagnetic sample shown a net magnetisation of 8 Am^{-1} when placed in an external magnetic field of 0.6T at a temperture of 4K . When the same is placed in an external magnetic field of 0.2T at a temeprature of 16K , the magnetisation will be

(A) $\frac{32}{3} \text{ Am}^{-1}$

(B) $\frac{2}{3} \text{ Am}^{-1}$

(C) 6 Am^{-1}

(D) 2.4 Am^{-1}

Ans. B

Sol. $\frac{I}{H} = \frac{C}{T}$
 $\frac{8 \times 0.2}{0.6 \times I} = \frac{16}{4}$
 $I = 2/3 \text{ Am}^{-1}$

2. The phenomena involved in the reflection of radiowaves by ionosphere is similar to

(A) reflection of light by a plane mirror

(B) total internal reflection of light in air during a mirage

(C) Dispersion of light by water molecules during the formation of a rainbow.

(D) Scattering of light by the particles of air.

Ans. B

Sol. Based on internal reflection.

3. An electron (mass m), debroglie wavelength λ_0 with an initial velcoity $v = v_0 \hat{i}$ ($v_0 > 0$) is in an electric field $E = -E_0 \hat{i}$ ($E_0 = \text{constant} > 0$). It's de Broglie wavelength at time t is given by

(A) $\frac{\lambda_0}{\left(1 + \frac{eE_0 t}{m v_0}\right)}$

(B) $\lambda_0 \left(1 + \frac{eE_0 t}{mv_0} \right)$

(C) λ_0

(D) $\lambda_0 t$

Ans. A

Sol. $\lambda = \frac{h}{mv} = \frac{\lambda_0}{\left(1 + \frac{eE_0 t}{m v_0} \right)}$

4. Fusion processes, like combining two deuterons to form a He nucleus are impossible at ordinary temperature and pressure. The reasons for this can be traced to the fact:

(A) nuclear forces have short range

(B) nuclear forces are charge independent

(C) the original nuclei must be completely ionized before fusion can take place

(D) The original nuclei must first break up before combining with each other.

Ans. A

Sol. At ordinary temperature and pressure, the separation will be out of nuclear force range.

5. A linearly polarized electromagnetic wave given as $\vec{E} = E_0 \hat{i} \cos(kz - \omega t)$ is incident normally on a perfectly reflecting infinite wall at $z = a$. Assuming that the material of the wall is optically inactive, the reflected wave will be given as

(A) $\vec{E}_r = -E_0 \hat{i} \cos(kz - \omega t)$

(B) $\vec{E}_r = E_0 \hat{i} \cos(kz + \omega t)$

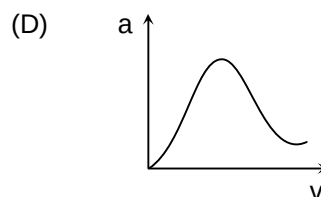
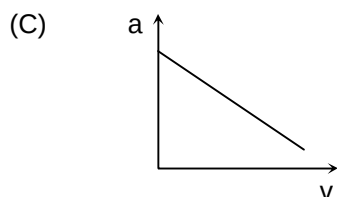
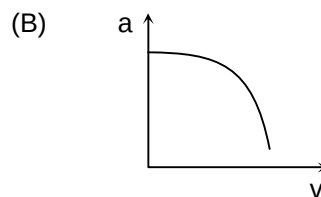
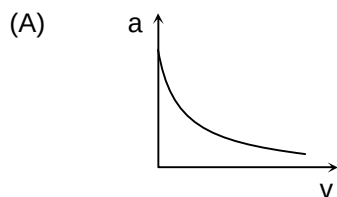
(C) $\vec{E}_r = -E_0 \hat{i} \cos(kz + \omega t)$

(D) $\vec{E}_r = E_0 \hat{i} \cos(kz - \omega t)$

Ans. B

Sol. From optically inactive surface there will be no phase change.

6. A small sphere is dropped in a viscous liquid, motion of sphere follows Stoke's Law. It attains terminal velocity after some time. The acceleration of ball varies with velocity as



Ans. C

Sol. $\frac{dv}{v_0 - v} = kdt$
 $v = v_0(1 - e^{-kt})$
 $\frac{dv}{dt} = kv_0 e^{-kt}$
 $v = v_0 - \frac{a}{k}$

7. A current I passes through an equilateral triangle of side 'a' the magnetic flux density at centroid of triangle will be

(A) $\frac{12\mu_0 I}{4\pi a}$

(B) $\frac{6\mu_0 I}{4\pi a}$

(C) $\frac{18\mu_0 I}{4\pi a}$

(D) $\frac{24\mu_0 I}{4\pi a}$

Ans. C

Sol. $B_1 = \frac{\mu_0 I}{4\pi \frac{a}{2\sqrt{3}}} 2 \cos 60^\circ = \frac{\mu_0 I}{4\pi a} \cdot \frac{\sqrt{3} \times \sqrt{3} \times 2}{4}$
 $B_1 = \frac{3}{4} \frac{\mu_0 I}{\pi a} \times 2 = \frac{6\mu_0 I}{4\pi a}$
 $B = 3B_1 \Rightarrow \frac{18\mu_0 I}{4\pi a}$

8. A monochromatic light of wavelength λ is incident on an isolated metal sphere of radius a the threshold wave length is λ_0 which is larger than λ . The number of photoelectrons emitted before photoelectron emission stops is

(A) $4\pi\epsilon_0 \frac{ahc}{e^2} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$

(B) $\frac{ahc}{4\pi\epsilon_0 e^2} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$

(C) $\frac{4\pi\epsilon_0 e^2}{ahc} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$

(D) $\frac{e^2}{4\pi\epsilon_0 ahc} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$

Ans. A

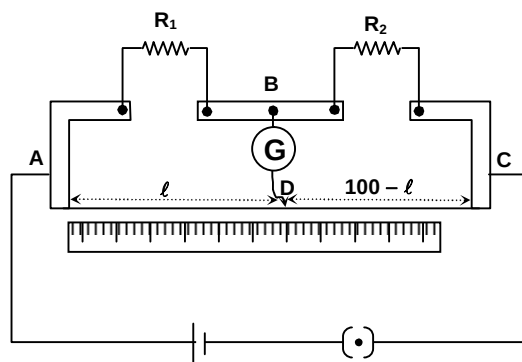
Sol.
$$eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$V = \frac{hc}{e} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

$$Q = 4\pi\epsilon_0 aV$$

$$n = \frac{Q}{e} = 4\pi\epsilon_0 \frac{ahc}{e^2} \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

9. In a meter bridge the point D is a neutral point as shown in the figure.

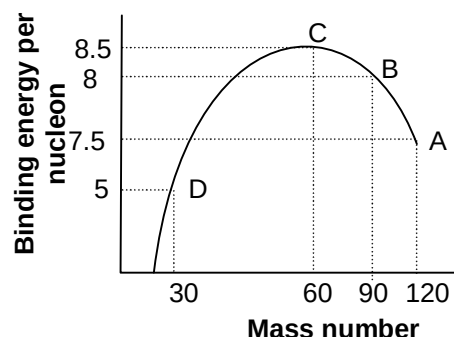


- (A) The meter bridge can have one more neutral point for this set of resistances.
- (B) When the jockey contacts a point on meter wire left of D. Current flow to B from the wire.
- (C) When the jockey contacts a point on the meter wire to the right of D. Current flows from B to the wire through galvanometer.
- (D) When R is increased, the neutral point shifts to left.

Ans. C

Sol. $\frac{R_1}{\ell} = \frac{R_2}{100 - \ell}$

10. Binding energy per nucleon versus mass number curve for nuclear is shown in the figure. A, B, C and D are four nucleus indicated on the curve. The process which release energy



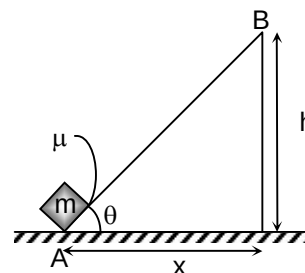
- (A) $C = D + D$
 (B) $A = B + D$
 (C) $A = 2C$
 (D) $B = C + D$

Ans. C

Sol. If Binding energy of products is greater than the Binding energy of reactants the energy will be released.

11. A particle of mass m is moved from A to B slowly along the inclined plane. θ is angle of inclined plane and μ is the coefficient of friction.

- (I) work done by gravity will be $-mgh$
 (II) work done by friction will be $-\mu mgx$
 (III) work done by external force will be $(mgh + \mu mgx)$
 (IV) work done by friction will be $\mu mg\sqrt{h^2 + x^2}$



- (A) (I), (II), (III) are correct
 (B) (I), (II), (III), (IV) are correct
 (C) (I) is correct
 (D) (I), (II) are correct

Ans. C

Sol. Work done by friction and external force will depend on the direction of external force.

12. A spherical black body with radius R radiates 500 watt power at 500 K. If radius is doubled and temperature is halved the power radiated will be

(A) 125 watt
 (B) 2000 watt
 (C) 500 watt
 (D) 4000 watt

Ans. A

Sol. $E = 4\pi R^2 \sigma T^4$

$$\frac{E_1}{E_2} = \frac{R_1^2 T_1^4}{R_2^2 T_2^4}$$

13. A particle m is moving inside of a heavy tube with velocity v_0 a heavier particle of mass M ($M \gg m$) also moving with velocity v_0 towards m . Initial separation of M from the end of tube is $3v_0$. The kinetic energy of particle m when separation between heavy particle and end of tube is v_0 will be (all the collisions are elastic in nature)



(A) $\frac{1}{2}mv_0^2$
 (B) $\frac{9}{2}mv_0^2$
 (C) $\frac{8}{2}mv_0^2$
 (D) $\frac{3}{2}mv_0^2$

Ans. B

Sol. Only one collision will take place after collision velocity of m will be $3v_0$

14. A parallel glass slab of refractive index ($n = \sqrt{3}$) is of thickness $t = 10$ cm. then the angle of incidence at first surface of air glass interface so that the reflected and emergent ray are perpendicular to each other.

(A) 30°
 (B) 60°
 (C) 45°
 (D) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Ans. C

Sol. Incident ray and emergent ray are parallel when passes through a glass slab. So, reflected and emergent ray will be perpendicular when $i = 45^\circ$.

15. The acceleration due to gravity on the surface of earth is 6 times that on the surface of moon and radius of moon is one fourth of the radius of earth the ratio of escape velocities on earth to moon will be

(A) $\sqrt{\frac{3}{2}}$

(B) 3

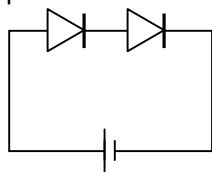
(C) $2\sqrt{6}$

(D) $\frac{\sqrt{3}}{2}$

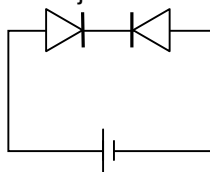
Ans. C

Sol.
$$v_{es} = \frac{\sqrt{2g_e R_e}}{\sqrt{2G_m R_m}} = \sqrt{\frac{6g_e}{g_e} \frac{4R_e}{R_e}} = 2\sqrt{6}$$

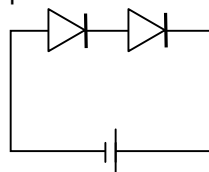
16. Two identical P-N junction diode may be connected in series with a battery in three ways. The potential difference across two P-N junction diode are equal in



(I)



(II)



(III)

(A) Circuit (I) and circuit (II)

(B) Circuit (II) and circuit (III)

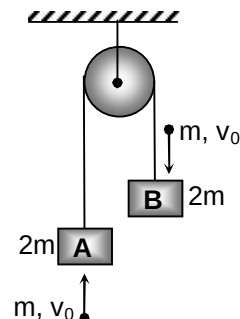
(C) Circuit (III) and circuit (I)

(D) Circuit (I) only

Ans. C

Sol. In circuit (I) both are forward bias. In circuit (II) both are reverse bias.

17. Two particles each of mass m each moving with velocity v_0 strikes to masses $2m$ and sticks to them simultaneously as shown in the figure. The velocity of mass B just after collision will be

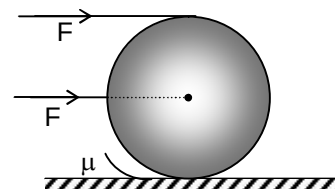


- (A) v_0
 (B) $\frac{v_0}{3}$
 (C) $\frac{v_0}{5}$
 (D) $\frac{v_0}{2}$

Ans. B

Sol. $mv_0 + mv_0 = 6mv$

18. A solid cylinder of mass M and radius R is pushed by two forces. F each as shown in the figure coefficient of friction is μ between the cylinder and the ground as shown in the figure. Choose the correct option.



- (A) The cylinder will roll without slipping.
 (B) The cylinder will roll with sliding.
 (C) The friction will act on cylinder towards right
 (D) The friction will act on cylinder towards left

Ans. A

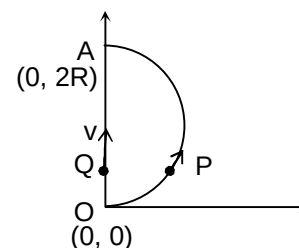
Sol. $2F = Ma_{cm}$... (i)

$$FR = \frac{MR^2}{2} \alpha \quad \dots (ii)$$

$\therefore a_{cm} = R\alpha$ without friction

So, friction is zero and cylinder will roll without slipping.

19. Two particle P and Q starts from origin P along a semicircular path of radius R with constant speed and Q moves along y-axis with constant speed v such that motion of P observed by Q is always one dimensional. The average speed of P for semicircular path will be



- (A) πv
 (B) $\frac{\pi v}{2}$
 (C) $\frac{v}{2}$
 (D) v

Ans. B

Sol. $t = \frac{2R}{v}$

Distance πR

$$v_{\text{ave}} = \frac{\pi R}{2R} v = \frac{\pi v}{2}$$

20. When a nucleus in an atom undergoes a radioactive decay, the electronic energy levels of the atom.
- (A) do not change for any type of radioactivity.
 (B) change for α and β radioactivity but not for γ -radioactivity.
 (C) change for α -radioactivity but not for others.
 (D) change for β -radioactivity but not for others.

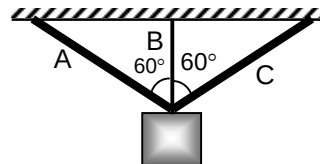
Ans. B

Sol. By α emission mass number reduces by 4 and atomic number reduces by 2. and by β decay atomic number increases by 1.

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

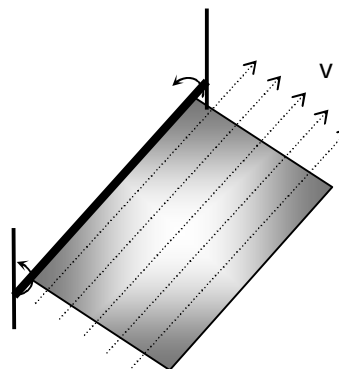
21. A mass $m = 10$ kg is suspended by three wires A, B and C of same material. Radius of wire A and C are double that of B. Calculate tension (in newton) in wire 'B' (elongation in wire is very small very small)



Ans. 00050.000

Sol. $T_A = T_C = T$
 $2T \cos \theta + T_B = mg$
 $2k_A x' \cos \theta + k_B x = mg$
 $2k_A x \cos^2 \theta + k_B x = mg$
 $k_A = \frac{YA \cos \theta}{\ell}, \quad k_B = \frac{YA}{\ell}$
 $k_A = 4 \cos \theta k_B$
 $4k_B x \cos^3 \theta + k_B x = mg$
 $T = \frac{mg}{1 + 8 \cos^3 \theta} = \frac{100}{2} = 50 \text{ N}$

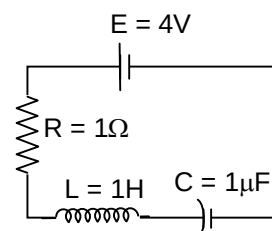
22. A thin plate of area $A = 100 \text{ cm}^2$, mass 260 gm hinged along one its side. Now the air is blown parallel to the upper surface in such away the plate remains horizontal. Calculate the velocity (in m/s) of air. (Given: density of air 1.3 kg/m^3 and $g = 10 \text{ m/s}^2$)



Ans. 00020.00

Sol. $mg \times \frac{a}{2} = (P_0 - P) \frac{a}{2} A$
 $mg = (P_0 - P_0 + \frac{1}{2} \rho v^2) A$
 $\sqrt{\frac{2 \times 0.26 \times 10}{1.3 \times 100 \times 10^{-4}}} = 20 \text{ m/s}$

23. The current in the given circuit is increasing with at a rate $a = 4$ amp/s. The charge (in μC) on the capacitor at an instant when the current in the circuit is 2 amp is $K\mu\text{C}$. Find the value of K .



Ans. 00002.00

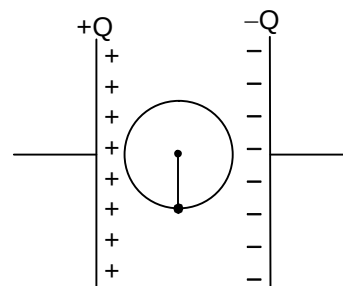
Sol. $iR + \frac{L di}{dt} + \frac{q}{C} = E$
 $2R + 1 \times 4 + \frac{K}{1\mu\text{F}} = 4$
 $\Rightarrow K = 2\mu\text{C}$

24. A progressive wave of amplitude 2.5 cm, wavelength 30 cm and frequency 80 Hz is superposed on a similar wave travelling the opposite direction to form a standing wave, the particle in the medium has maximum velocity v_0 m/s. Calculate the velocity v_0 in m/s. (Take $\pi = 3.14$)

Ans. 00027.12

Sol. $v = 2A\omega$

25. A particle of mass m charge q is in equilibrium inside of an parallel plate capacitor having charge Q and area of plate A charge q is suspended by a string of length $\ell = \sqrt{2}m$. If $\frac{Qq}{A\epsilon_0} = mg$, calculate the minimum velocity given to the particle so that it completes vertical circle. (Take $g = 10 \text{ m/s}^2$)



Ans. 00010.00

Sol. $v_{\min} = \sqrt{5g\sqrt{2}\ell} = \sqrt{5 \times 10 \times 2} = 10 \text{ m/s}$

26. The focal length of the objective lens and eyepiece lens of a compound microscope are 4 cm and 5 cm respectively. If an object is placed at a distance of 6 cm in front of objective lens what is the magnification produced by microscope. (least distance of distinct vision = 25 cm)

Ans. 00012.00

Sol. $u_0 = 6 \text{ cm}, f_0 = 4 \text{ cm}, v_0 = 12 \text{ cm}$
 $M = \frac{v_0}{u_0} \left(1 + \frac{D}{f_e} \right)$
 $M = \frac{12}{6} \left(1 + \frac{25}{5} \right) = 12$

27. A metal ring of radius $r = 1\text{m}$ with its plane normal to uniform magnetic field $B = 4\text{T}$ carries a current of 10 A . The tension developed in the ring (in newton) is

Ans. 00040.00

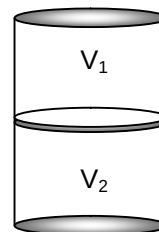
Sol. $2T \sin \frac{d\theta}{2} = iBd\ell$
 $T = iBr = 10 \times 1 \times 4 = 40\text{ N}$

28. A diatomic gas ($\gamma = 1.4$) is expanded isobarically if work done by gas is 200 J . Calculate the heat given to the gas.

Ans. 00700.00

Sol. $\Delta Q = nC_p \Delta T$
 $nR \Delta T = 200\text{ J}$
 $nC_v \Delta T = \frac{nR \Delta T}{\gamma - 1} = \frac{200}{0.4} = 500$
 $\Delta Q = 700\text{ J}$

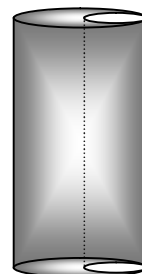
29. A freely moving piston divides a vertical cylinder closed at both ends into two parts each containing 1 mole of ideal gas. In equilibrium at 300°K volume of upper part is 4 times greater than lower part. Calculate the temperature (in kelvin scale) when ratio of these volume is 2.



Ans. 00750.00

Sol. $P_2 - P_1 = \frac{mg}{A}$
 $\frac{nRT_1}{V_2} - \frac{nRT_1}{4V_2} = \frac{mg}{A}$
 $\frac{nRT_2}{V'_2} - \frac{nRT_2}{2V'_2} = \frac{mg}{A}$
 $5V_2 = V$ and $3V'_2 = V$
 Solving $T_2 = \frac{5T_1}{2} = 750$

30. A cylindrical wire of radius R is having a cylindrical cavity of radius $R/2$ the current density in wire is J . A charge q is moving inside of cylindrical smooth cavity with constant speed v . Such that the magnetic field on the axis of cavity is zero. The speed of charge particle. (Take $J = 2\text{ amp/m}^2$. $R = 0.5\text{ m}$ and $q = \frac{\pi}{64}\text{ coulomb}$)



Ans. 00004.00

Sol. $B_1 \cdot 2\pi \frac{R}{2} = \mu_0 \frac{I}{\pi R^2} \frac{\pi R^2}{4}$

$$B_1 = \frac{\mu_0}{4} \frac{\pi R^2}{\pi R} J = \frac{\mu_0 R}{4} J$$

$$B_2 = \frac{\mu_0}{2} \frac{R}{2} \cdot \frac{qV}{2\pi R} = \frac{\mu_0 qV}{\pi R^2}$$

$$\frac{\mu_0 R}{4} J = \frac{\mu_0 qV}{\pi R^2}$$

$$V = \frac{\pi R^3}{4q} J = \frac{\pi \times 64 \times 1}{4\pi \times 8} \times 2 = 4$$

Chemistry

PART – B

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following statement is not true?

- (A) The energy of emitted electron in an photoelectric effect is independent of intensity of light.
- (B) A catalyst that enhance the rate, lower the energy of activation.
- (C) A zero order reaction can be an elementary reaction.
- (D) A cubic crystal possess 23 element of symmetry.

Ans. C

Sol. A zero order reaction must be a complex reaction.

32. Equal mole of two liquid A and B mixed to form an ideal solution, whose $\frac{P_A^0}{P_B^0}$ value is $\frac{3}{2}$. What will be the ratio of mole fraction of A and B in distillate after three repeated distillations?

- (A) $\frac{9}{4}$
- (B) $\frac{4}{9}$
- (C) $\frac{27}{8}$
- (D) $\frac{3}{2}$

Ans. C

Sol. Initially $\frac{X_A}{X_B} = 1$

Ratio of mole fraction of A and B after first distillation in distillate $\frac{X'_A}{X'_B} = \frac{P_A^0 X_A}{P_B^0 X_B} = \frac{3}{2}$

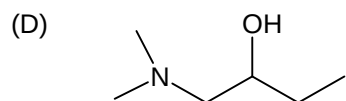
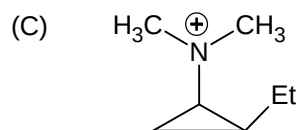
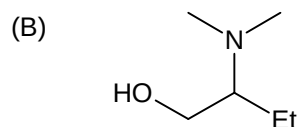
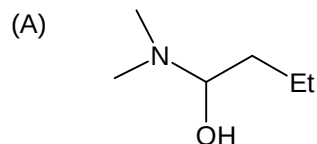
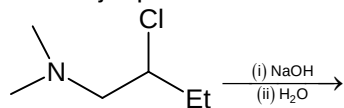
Ratio of mole fraction of A and B after second distillation in distillate $\frac{X''_A}{X''_B} = \frac{P_A^0}{P_B^0} \times \frac{X'_A}{X'_B}$
 $= \frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$

Ratio of mole fraction of A and B after third distillation in distillate

$$\frac{X_A'''}{X_B'''} = \frac{P_A^0}{P_B^0} \times \frac{X_A''}{X_B''}$$

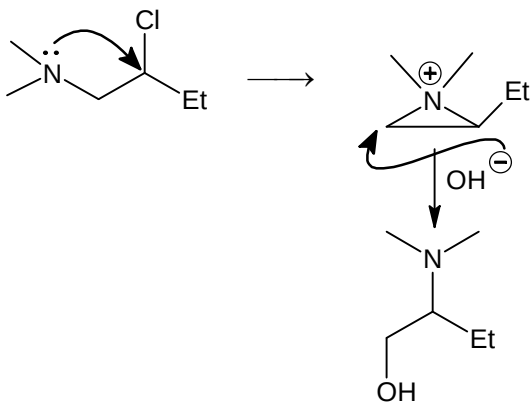
$$= \frac{3}{2} \times \frac{9}{4} = \frac{27}{8}$$

33. The major product of the following reaction is

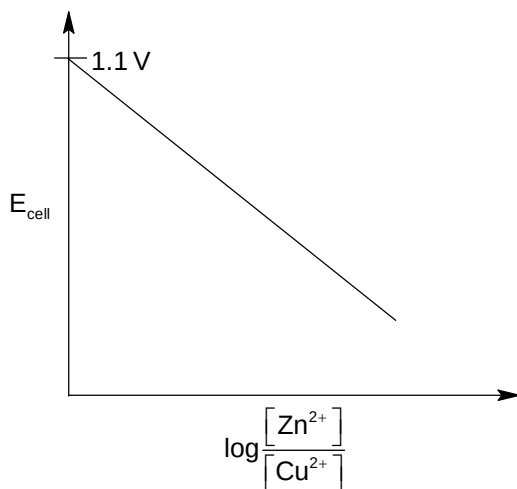
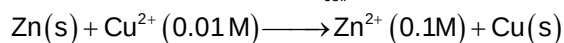


Ans. B

Sol.



34. What will be the value of E_{cell} for the following cell reaction at room temperature?



- (A) 1.1295 V
 (B) 1.0705 V
 (C) 1.159 V
 (D) 1.0305 V

Ans. B

Sol. Cell
 $\text{Zn} | \text{Zn}^{2+} || \text{Cu}^{2+} | \text{Cu}$

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \\ &= 1.1 - \frac{0.059}{2} \log \frac{0.1}{0.01} \\ &= 1.1 - \frac{0.059}{2} = 1.0705 \text{ V} \end{aligned}$$

35. Choose the incorrect statement regarding carbon monoxide?

- (A) It is a neutral oxide.
 (B) It turns PdCl_2 solution black.
 (C) It burns with blue flame.
 (D) It oxidises Fe to Fe_2O_3 .

Ans. D

Sol. It reduces Fe_2O_3 to Fe.

36. The correct statement among following is

- (A) B_2Cl_4 is planar in gaseous state
- (B) B_2Cl_4 is staggered in solid state
- (C) Thermal stability of B_2Cl_4 is Less than B_2F_4
- (D) Like B_2H_6 there is no B – B bond in B_2Cl_4

Ans. C

Sol. B_2Cl_4 (s) – planar

B_2Cl_4 (g) – staggered

B_2F_4 is more stable as F has greater tendency of do back bonding with boron.

37. The incorrect option is

- (A) $HCl < HBr < HI < HF$ (Boiling point order)
- (B) $H_2S < H_2Se < H_2Te < H_2O$ (Melting point order)
- (C) $H_2O < H_2S < H_2Se < H_2Te$ (Acidic strength)
- (D) $NH_3 < PH_3 < AsH_3 < SbH_3$ (Heat of vaporization)

Ans. D

Sol. Correct order of heat of vaporization

$PH_3 < AsH_3 < SbH_3 < NH_3$ due to H-bonding.

38. Which of the following drug is not used for hyper acidity?

- (A) Ranitidine
- (B) Cimetidine
- (C) Aluminium hydroxide gel
- (D) Codeine

Ans. D

Sol. Codeine is an analgesic drug.

39. Which of the following statement is not true about photochemical smog?

- (A) Ozone is a primary precursor of the photochemical smog.
- (B) It damage plant life

- (C) Ozone, nitric oxide, acrolein, formaldehyde etc. are common components of photochemical smog.
- (D) It can cause eye irritation

Ans. A

Sol. Ozone is a secondary precursor of photochemical smog.

40. Which of the following gas has its most probable speed at 327°C equal to its RMS speed at 127°C?
- (A) He
- (B) H₂
- (C) Ar
- (D) All of these

Ans. D

Sol.
$$V_{m.p.} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2R \times 600}{M}} = \sqrt{\frac{1200R}{M}}$$

$$V_{RMS} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3R \times 400}{M}} = \sqrt{\frac{1200R}{M}}$$

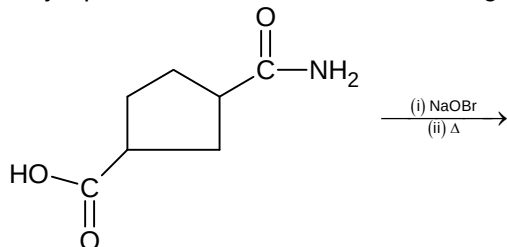
$$\therefore V_{m.p.}(600K) = V_{RMS}(400K) \text{ for all gases.}$$

41. Choose the incorrect statement.
- (A) I⁻ is better nucleophile than F⁻ in CH₃OH
- (B) F⁻ is better nucleophile than I⁻ in DMSO
- (C) I⁻ is better nucleophile than F⁻ in hexane
- (D) F⁻ is a good nucleophile in gas phase reaction

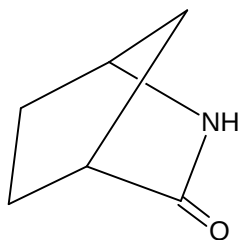
Ans. C

Sol. F⁻ is better nucleophile in polar aprotic and non-polar solvent as well as in gas phase reaction.

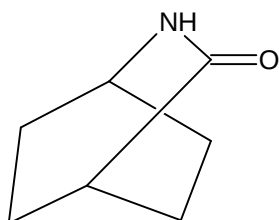
42. Major product obtained from the following reaction is



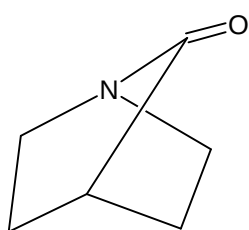
(A)



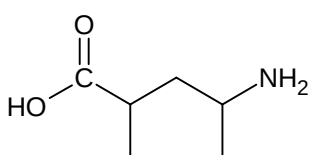
(B)



(C)

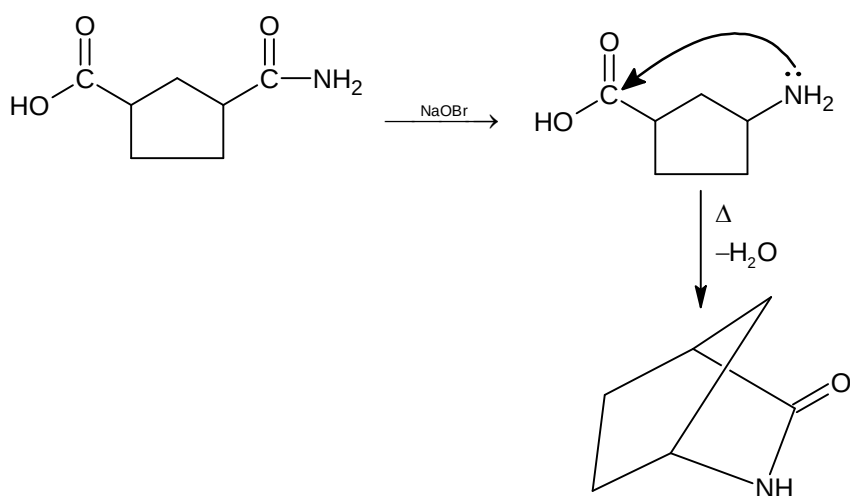


(D)



Ans. A

Sol.



43. Which of the following is an amphoteric oxide?

- (A) CO_2
- (B) N_2O
- (C) Al_2O_3
- (D) MgO

Ans. C

Sol. CO_2 - acidic oxide
 N_2O - neutral oxide
 Al_2O_3 - amphoteric oxide
 MgO - basic oxide

44. Which of the following complex is sp^3 hybridized as well as attracted by magnet?

- (A) $[\text{Ni}(\text{CN})_4]^{2-}$
- (B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- (C) $[\text{NiCl}_4]^{2-}$
- (D) $[\text{Ni}(\text{CO})_4]$

Ans. C

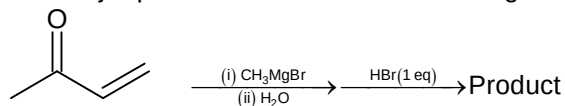
Sol. $[\text{Ni}(\text{CN})_4]^{2-} \rightarrow$ Square planar, diamagnetic

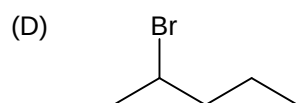
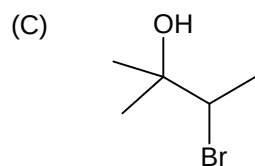
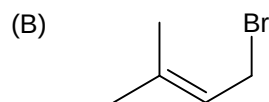
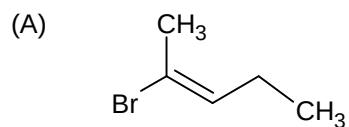
$[\text{Cu}(\text{NH}_3)_4]^{2+} \rightarrow$ Square planar, paramagnetic

$[\text{Ni}(\text{CO})_4]$ Tetrahedral, diamagnetic

$[\text{NiCl}_4]^{2-} \rightarrow$ Tetrahedral, paramagnetic
 $\downarrow$
 Attracted by magnet

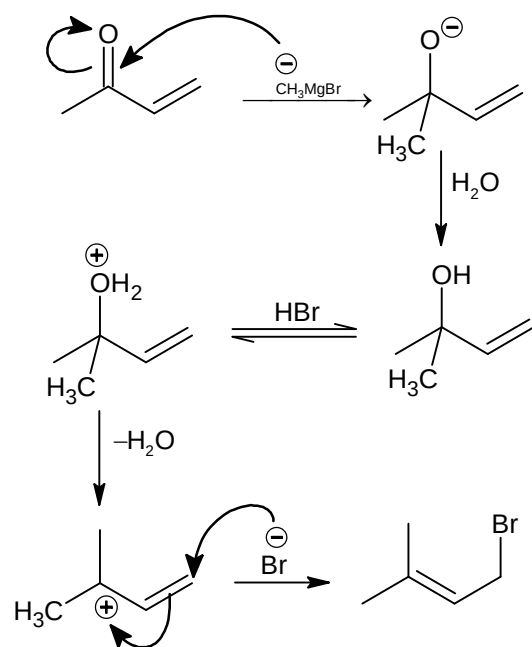
45. The major product obtained in the following reaction is



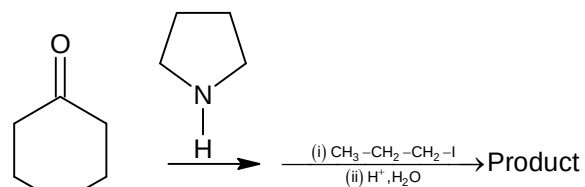


Ans. B

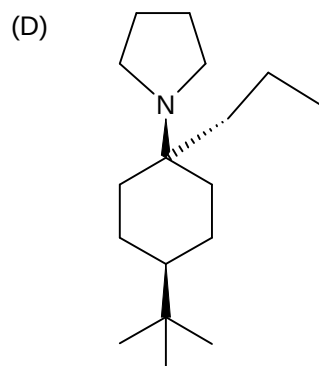
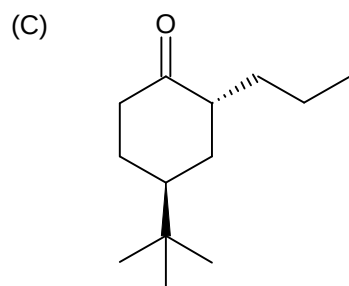
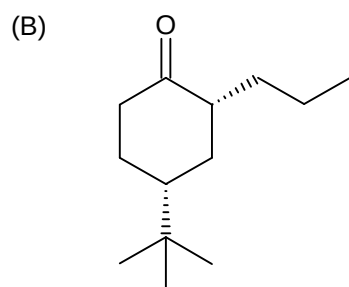
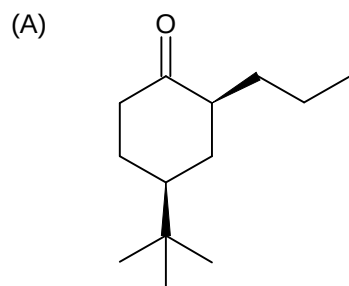
Sol.



46.

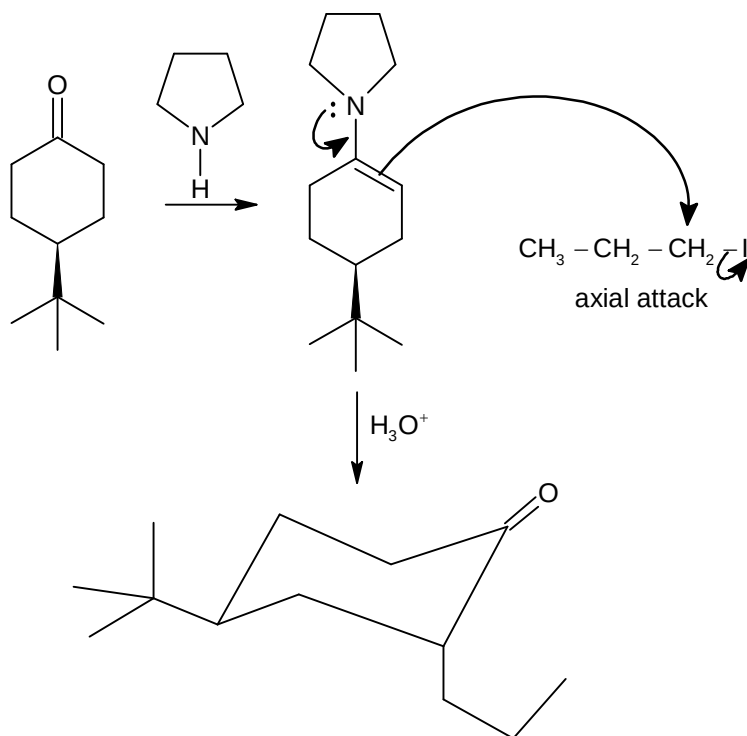


Major product is:



Ans. C

Sol.



47. Choose the correct option for the following statements

- I. Equilibrium constant tells us about how far a reaction would go.
- II. Doubling the stoichiometric coefficient of a chemical equation double the numerical value of equilibrium constant.
- III. For a gaseous reaction with $\Delta n_g = +ve$, addition of inert gas of constant pressure shift the equilibrium towards left.
- IV. For a gaseous reaction with $\Delta n_g = -ve$, increase of pressure causes shift of equilibrium towards right.

- (A) I and II are correct.
- (B) II and IV are correct.
- (C) Only I is correct.
- (D) II and III are incorrect.

Ans. D

Sol. Factual

48. Choose the correct statement

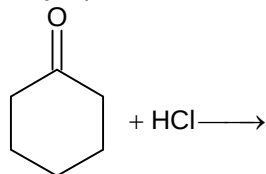
- (A) When an ideal gas is heated from 20°C to 40°C at constant pressure its volume doubles.
- (B) A catalyst can increase the rate of such a reaction which is thermodynamically not feasible.

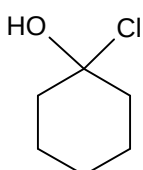
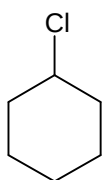
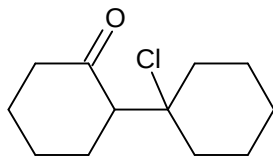
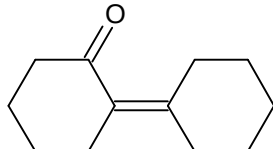
- (C) Particles of solute in a true solution may scatter visible light.
(D) Emulsions are lyophobic colloids.

Ans. D

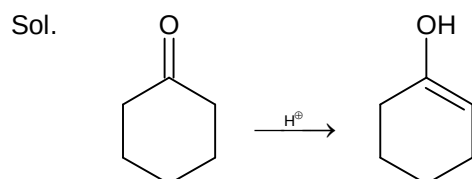
Sol. Factual.

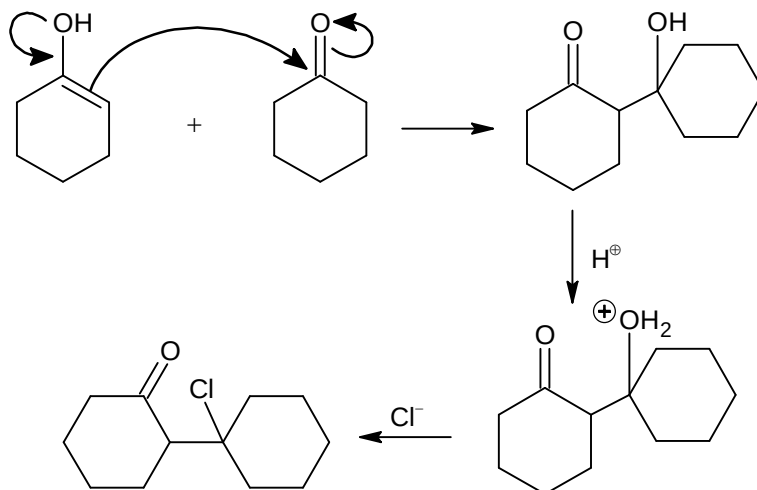
49. Major product obtained for following reaction is



- (A)  (A cyclohexane ring with a hydroxyl group and a chlorine atom on the same carbon atom.)
- (B)  (A cyclohexane ring with a chlorine atom attached to one of the carbons.)
- (C)  (Two cyclohexane rings connected by a single bond. One ring has a ketone group, and the other ring has a chlorine atom on the carbon adjacent to the bond.)
- (D)  (Two cyclohexane rings connected by a double bond. One ring has a ketone group, and the other ring is attached to the double bond via a methylene group.)

Ans. C



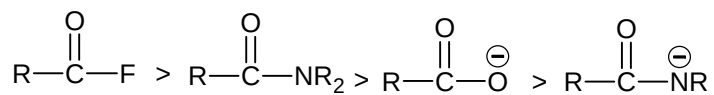


50. Least reactive species among the following towards nucleophilic addition is

- (A) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{F}$
- (B) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NR}^{\ominus}$
- (C) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}^{\ominus}$
- (D) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NR}_2$

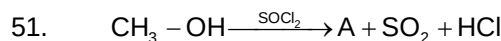
Ans. B

Sol. Order of electron donation by resonance is $\text{NR}^{\ominus} > \text{O}^{\ominus} > \text{NR}_2 > \text{F}$
 $\therefore$ Reactivity order



SECTION – B
(Numerical Answer Type)

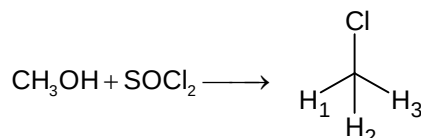
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).



Total number of plane in product 'A' which contains maximum number of atoms is?

Ans. 00010.00

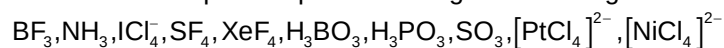
Sol.



Planes which contain maximum atom (3) are

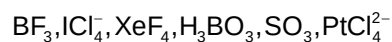
H_1CH_2 , H_2CH_3 , H_1CH_3 , H_1CCl , H_2CCl , H_3CCl , $\text{H}_1\text{H}_2\text{Cl}$, $\text{H}_1\text{H}_3\text{Cl}$, $\text{H}_2\text{H}_3\text{Cl}$, $\text{H}_1\text{H}_2\text{H}_3$
= 10.

52. Total number of planar species among the following is



Ans. 00006.00

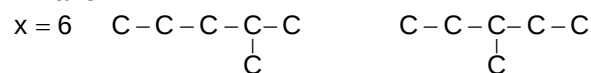
Sol. Planar molecules/ions are



53. If the minimum number of carbon required for an alkane to show position isomer is 'x' and minimum number of carbon required for an alkyne to show chain isomer is 'y', the value of x + y is:

Ans. 00011.00

Sol. Alkane



Position isomers

Alkyne



Chain isomers

$\therefore x + y = 11$

54. What will be the value of change in entropy in $\text{JK}^{-1}\text{mol}^{-1}$ when 0.1 mole of an ideal gas expanded reversibly two times of its initial volume at 25°C . ($\ln 2 = 0.69$).

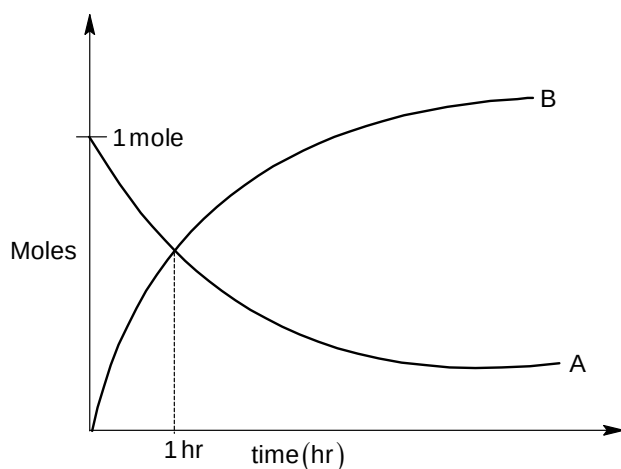
Ans. 00000.57

Sol.
$$\Delta S = \frac{q}{T} = \frac{nRT \ln \frac{V_2}{V_1}}{T}$$

$$= 0.1 \times 8.314 \times \ln(2)$$

$$= 0.5736 \text{ JK}^{-1}\text{mol}^{-1}$$

55. For a first order reaction $A \longrightarrow 2B$ (graph shown below) the mole of B formed after 2 hour is $\frac{x}{9}$. The value of x is:



Ans. 00010.00

Sol.
$$A \longrightarrow B$$

$$1 \quad 0$$
 at 1 hr $1-a \quad 2a$
 at $t = 1 \text{ hr}$ $1-a = 2a$

$$\therefore a = \frac{1}{3}$$
 at $t = 2 \text{ hr}$ $1-b \quad 2b$

$$k = \frac{2.303}{1} \log \frac{1}{1-\frac{1}{3}} = 2.303 \log \left(\frac{3}{2} \right)$$
 At $t = 2 \text{ hr}$

$$k = \frac{2.303}{2} \log \left(\frac{1}{1-b} \right) = 2.303 \log \left(\frac{3}{2} \right)$$

$$\therefore \log \frac{1}{1-b} = \log \left(\frac{3}{2} \right)^2$$

$$\frac{1}{1-b} = \frac{9}{4}$$

$$b = \frac{5}{9}$$

$$\therefore [B] = 2b = \frac{10}{9}$$

$$\frac{10}{9} = \frac{x}{9}$$

$$\therefore x = 10$$

56. If the wavelengths emitted during electronic transition between two levels of He^+ ion whose sum is 5 and difference is 1 is $x \text{ cm}^{-1}$ and two levels of H atom whose sum is 4 and difference is 2 is $y \text{ cm}^{-1}$, then value of x/y is:

Ans. 00001.60

Sol. For He^+ $n_1 + n_2 = 5$
 $n_2 - n_1 = 1$
 $\therefore n_1 = 2, n_2 = 3$

$$\frac{1}{\lambda_{\text{He}^+}} = R_H (2)^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$= 4R_H \times \frac{5}{36} = \frac{5}{9} R_H$$

$$x = \lambda_{\text{He}^+} = \frac{9}{5R_H} \text{ cm}^{-1}$$

For H – atom

$$n_1 + n_2 = 4$$

$$n_2 - n_1 = 2$$

$$\therefore n_1 = 1, n_2 = 3$$

$$\frac{1}{\lambda_H} = R_H (1)^2 \left\{ \frac{1}{1^2} - \frac{1}{3^2} \right\}$$

$$= \frac{8}{9} R_H$$

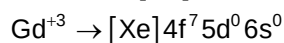
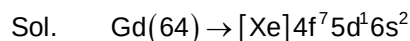
$$\therefore y = \lambda_H = \frac{9}{8R_H}$$

$$\therefore \frac{x}{y} = \frac{\frac{9}{5R_H}}{\frac{9}{8R_H}}$$

$$= \frac{8}{5} = 1.6$$

57. Number of unpaired electron in Gd^{+3} ion is (Atomic number of Gd = 64).

Ans. 00007.00



58. In 100 ml, 0.1 M CH_3COOH solution, 0.1 M NaOH is added till pH of mixture become 5. The volume of NaOH (ml) used is

Given: $\text{pK}_{\text{a}}(\text{CH}_3\text{COOH}) = 4.82$, $\log\left(\frac{3}{2}\right) = 0.18$.

Ans. 00060.00



$$\begin{array}{cccc} 100 \times 0.1 & V \times 0.1 & 0 & 0 \\ 10 - 0.1V & 0 & 0.1V & \end{array}$$

$$\text{pH} = \text{pK}_{\text{a}} + \log \frac{[\text{CH}_3\text{COONa}]}{[\text{CH}_3\text{COOH}]}$$

$$5 = 4.82 + \log \frac{0.1V}{10 - 0.1V}$$

$$0.18 = \log \left(\frac{0.1V}{10 - 0.1V} \right)$$

$$\log \left(\frac{3}{2} \right) = \log \left(\frac{0.1V}{10 - 0.1V} \right)$$

$$\therefore V = 60 \text{ ml}$$

59. What percentage of octahedral void is occupied by Fe^{+3} ion in Fe_3O_4 .

Ans. 00025.00

Sol. Fe_3O_4 is an inverse spinel in which out of 4 octahedral voids. One octahedral void occupied by Fe^{+3} and one occupied by Fe^{+2} .

$\therefore$ % of octahedral voids occupied by Fe^{+3} is 25%.

60. A solution is prepared by mixing 10 ml 0.2 M of R enantiomer and 20 ml 0.15 M of S enantiomer of a compound, was found to have an observed specific rotation of $+5.5^\circ$. What will be the specific rotation of S enantiomer?

Ans. 00027.50

Sol. m.mol of R enantiomer = $10 \times 0.2 = 2 \text{ m mol}$

m.mol of S enantiomer = $20 \times 0.15 = 3 \text{ m mol}$

$\therefore$ 4 m. mol of racemic mixture forms and 1 m.mol of S-enantiomer will be left over.

$$\therefore \text{E.excess} = \frac{1}{5} \times 100 = 20\%$$

$$\% \text{E.E} = \frac{\text{observe sp. rotation}}{\text{sp. rotation of pure enantiomer}} \times 100$$

$$20 = \frac{+5.5}{\text{sp. rotation of S - enantiomer}} \times 100$$

Specific rotation of S-enantiomer = $+27.5^\circ$.

Mathematics

PART – C

SECTION – A

(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $\lim_{x \rightarrow 0} \frac{\sin(\alpha x - \ln(1 + \beta x))}{e^{3x} - 3x - 1} = 1$, then $\alpha + 2\beta$ is equal to

- (A) 1
- (B) 4
- (C) 9
- (D) 16

Ans. C

Sol. $\lim_{x \rightarrow 0} \frac{\sin(\alpha x - \ln(1 + \beta x))}{e^{3x} - 3x - 1} = \lim_{x \rightarrow 0} \frac{\left((\alpha - \beta)x + \frac{\beta^2 x^2}{2} - \frac{\beta^3 x^3}{3} + \dots \right)}{\frac{9x^2}{2} + \frac{27x^3}{3!} + \dots} = 1$

$\Rightarrow \alpha = \beta, \beta^2 = 9 \Rightarrow (\alpha, \beta) = (3, 3) \text{ or } (-3, -3)$

62. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln\left(1 + \frac{\sin x}{2}\right)}{\sin x} dx$ is equal to

- (A) $\frac{\pi}{6}$
- (B) $\frac{\pi^2}{6}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{\pi^2}{3}$

Ans. B

Sol. Let $f(b) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(1 + b \sin x)}{\sin x} dx, |b| < 1$

$$\Rightarrow f'(b) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{1+b \sin x}$$

$$\Rightarrow f(b) = \pi \sin^{-1}(b)$$

63. The value of $\sum_{k=8}^{23} (-1)^{k+1} {}^{31}C_k$ is equal to

- (A) ${}^{30}C_{14}$
 (B) $2 \cdot {}^{31}C_7$
 (C) $2 \cdot {}^{32}C_{21}$
 (D) none of these

Ans. D

Sol. $\sum_{k=8}^{23} (-1)^{k+1} {}^{31}C_k = \sum_{k=8}^{23} \left((-1)^{k-1} {}^{30}C_{k-1} - (-1)^k {}^{30}C_k \right) = {}^{30}C_{23} - {}^{30}C_7 = 0$

64. Let $\alpha = e^{i\frac{5\pi}{21}}$ and $z = \alpha - \alpha^2 + \alpha^3 - \alpha^4 + \dots + \alpha^9 - \alpha^{10}$. If $A = \text{Re}(z)$, then $|A|$ is equal to

- (A) $\frac{1}{4}$
 (B) $\frac{1}{2}$
 (C) $\frac{3}{4}$
 (D) 1

Ans. B

Sol. $\alpha^{21} = -1 \Rightarrow 1 - \alpha + \alpha^2 - \alpha^3 + \dots + \alpha^{20} = 0$
 $\Rightarrow |\text{Re}(z)| = \frac{1}{2}$

65. The value of $\left[2 \int_0^{1/2} \frac{dx}{\sqrt{1-x^3}} \right]$ is equal to (where $[.]$ denotes greatest integer function)

- (A) -1
 (B) 0
 (C) 1
 (D) none of these

Ans. C

Sol. $\int_0^{1/2} dx < \int_0^{1/2} \frac{dx}{\sqrt{1-x^3}} < \int_0^{1/2} \frac{dx}{\sqrt{1-x^2}} \Rightarrow \frac{1}{2} < \int_0^{1/2} \frac{dx}{\sqrt{1-x^3}} < \frac{\pi}{6}$

66. Let focus of a parabola be α, β such that the lines $x + y - 3 = 0$, $x - y + 3 = 0$, $x + 2y - 3 = 0$ and $x - 2y + 3 = 0$ are tangents to the parabola, then the value of $\alpha - \beta$ is equal to

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Ans. B

Sol. Focus of the parabola is $(0, -1)$

67. Let $f(x)$ be a quadratic function satisfying the equation $\int_0^x f(t) dt = xf(x) - \frac{4x^3}{3} + ax^2 \quad \forall x \in \mathbb{R}$ and $a \in \mathbb{R}$. Also $f(0) = a^2 - a - 6$, then number of integral values of a such that $f(x) = 0$ has real roots is/are

- (A) 5
- (B) 6
- (C) 7
- (D) 8

Ans. C

Sol. $f(x) = 2x^2 - 2ax + a^2 - a - 6$
For real roots, $a \in [1 - \sqrt{13}, 1 + \sqrt{13}]$

68. Let $S_1(\alpha, 0)$ and $S_2(-\alpha, 0)$ be the foci of the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$, ($\alpha > 0$). If $S = 0$ be the locus of the image of the focus S_1 about any tangent on the ellipse, then area bounded by $S = 0$ and $x + 3y + 1 = 0$ is equal to

- (A) 2π sq. units
- (B) 4π sq. units
- (C) 8π sq. units
- (D) 16π sq. units

Ans. C

Sol. Locus is $(x + 1)^2 + y^2 = 16$

69. Eccentricity of the hyperbola $2x^2 - 2y^2 - 3xy + 9x + 12y = 0$ is equal to

(A) $\sqrt{2}$

(B) $\sqrt{3}$

(C) 2

(D) $\sqrt{5}$

Ans. A

Sol. Angle between asymptotes is 90°

$\Rightarrow$ Eccentricity = $\sqrt{2}$

70. A tower subtends a right angle at the top of a pole and distance between tower and pole is 10 m, then minimum height of the tower is

(A) 5 m

(B) 10 m

(C) 20 m

(D) none of these

Ans. C

Sol. If angle of elevation of top of the pole from bottom of the tower is θ , then height of the tower is

$\frac{20}{\sin 2\theta}$

71. Let a population A has 100 observations 101, 102,, 200 and another population B has 100 observations 151, 152,, 250. If V_A and V_B represent the variances of the two population respectively, then $\frac{V_A}{V_B}$ is equal to

(A) $\frac{9}{4}$

(B) $\frac{4}{9}$

(C) $\frac{2}{3}$

(D) 1

Ans. D

Sol. Variance is independent of change of origin

$\Rightarrow V_A = V_B$

72. $\sim(p \vee q) \vee (\sim p \wedge q)$ is logically equivalent to

- (A) $\sim p$
- (B) p
- (C) q
- (D) $\sim q$

Ans. A

Sol. $\sim(p \vee q) \vee (\sim p \wedge q)$
 $\equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q)$
 $\equiv \sim p \wedge (\sim q \vee q)$
 $\equiv \sim p$

73. Let α, β be the roots of the equation $x^2 + 2x + 5 = 0$, then the value of $\frac{4\beta^3 - 2\beta^2}{\alpha^3 + \alpha}$ is equal to

- (A) 4
- (B) 5
- (C) 6
- (D) 7

Ans. B

Sol. $2\beta^3 - \beta^2 = 25$ and $\alpha^3 + \alpha = 10$

74. 96^{th} term of the sequence 2, 7, 13, 20, 28, is

- (A) 4924
- (B) 4942
- (C) 4992
- (D) none of these

Ans. B

Sol. $T_n = \frac{(n+3)(n+4)}{2} - 8$

75. Number of integral solutions of the equation $\sin^{-1}|\cos x| = \cos^{-1}|\sin x|$ in $[0, 4\pi]$ is equal to

- (A) 5
- (B) 7
- (C) 11
- (D) 13

Ans. D

Sol. $\sin^{-1}|\cos x| = \cos^{-1}|\sin x|$
 $\Rightarrow x \in \mathbb{R}$

76. The value(s) of 'a' for which $f(x) = ax^3 + (a + 6)x^2 + 4x + 7$ is monotonic $\forall x < 0$ is/are

- (A) $(-\infty, 2)$
- (B) $(0, \infty)$
- (C) $(-2, 3)$
- (D) none of these

Ans. D

Sol. $f'(x) = 3ax^2 + 2(a + 6)x + 4$
 For $f(x)$ to be monotonic $\forall x < 0$, $f'(x) = 0$ has both roots non-negative
 $\Rightarrow a > 0$

77. Area bounded by the curves $y^2 = 4x$, $y^2 = -4x$ and $y = 4$ is equal to

- (A) $\frac{8}{3}$ sq. units
- (B) $\frac{16}{3}$ sq. units
- (C) $\frac{32}{3}$ sq. units
- (D) $\frac{64}{3}$ sq. units

Ans. C

Sol. Required area = $2 \int_0^4 \frac{y^2}{4} dy$

78. Equation of plane containing (1, 1, 2) and which is at a maximum distance from (2, 2, 1) is

- (A) $x + y + z = 0$
- (B) $x - y - z = 0$
- (C) $x - y + z = 0$
- (D) $x + y - z = 0$

Ans. D

Sol. Equation of plane is $1 \cdot (x - 1) + 1 \cdot (y - 1) - 1 \cdot (z - 2) = 0$
 $\Rightarrow x + y - z = 0$

79. Let $A = \{-3, -2, -1, 0, 1, 2, 3\}$, then number of functions from A to A which are odd and $f(i) \neq i$ ($\forall i = -3, -2, -1, 1, 2, 3$) is/are

- (A) 126
- (B) 216
- (C) 261
- (D) 162

Ans. B

Sol. Number of required function is equal to $6 \times 6 \times 6 = 216$

80. Let R be the region containing all the points within or on the sides of an equilateral triangle. A point P is randomly selected from the region R , the probability that P lies near to internal angle bisector than any of the sides is equal to

- (A) $4 - 2\sqrt{3}$
- (B) $2 - \sqrt{3}$
- (C) $3\sqrt{3} - 4$
- (D) none of these

Ans. A

Sol. Area of favourable region = $\frac{\sqrt{3}a^2}{4} - \frac{3(2 - \sqrt{3})a^2}{4}$
 $\Rightarrow$ Required probability = $4 - 2\sqrt{3}$

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Number of divisors of the number $N = 2^3 \cdot 3^2 \cdot 5^4 \cdot 7^2$ which are of the form $(4k + 2)$, $k \in \text{Integer}$ is/are

Ans. 00045.00

Sol. Number of divisors of the form $4k + 2 = 1 \times 3 \times 5 \times 3 = 45$

82. Let $p(\alpha, \beta)$ be the circum-centre of the triangle formed by the lines $2x + 2y - 7 = 0$ and $2x^2 + 7xy - 2y^2 = 0$, then the value of $\alpha + \beta$ is equal to

Ans. 00003.50

Sol. $p(\alpha, \beta)$ lies on $2x + 2y - 7 = 0$

83. Let $\overrightarrow{OA} = \hat{i} + 2\hat{j} - \hat{k}$, $\overrightarrow{OB} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\overrightarrow{OC} = 3\hat{i} + 2\hat{j} + \hat{k}$ represent edges of the tetrahedron OABC, then length of perpendicular from vertex O on the base ABC is equal to

Ans. 00002.00

Sol. Let length of perpendicular = H

$$\Rightarrow \frac{1}{3} \times 1 \times H = \frac{2}{3} \Rightarrow H = 2$$

84. Order of the differential equation of the family of parabolas having fixed directrix and fixed axis is

Ans. 00001.00

Sol. Number of arbitrary constant in the equation of the family of parabolas = 1

85. Number of values of k for which $x^2 + y^2 - 4x - 4y + k = 0$ intersect $x + 2y = 0$ and $2x + y = 0$ at exactly 3 points is/are

Ans. 00001.00

Sol. (2, 2) lies on the angle bisector of the lines $x + 2y = 0$ and $2x + y = 0$

86. Let $\int \frac{(3+4x)\sqrt{x}dx}{2\sqrt{x+1}} = x^{3/2}f(x) + c$, then the value of f(3) is equal to

Ans. 00002.00

Sol. $f(x) = \sqrt{x+1}$

87. Let A be a 3×3 matrix such that $\text{Adj}(A) = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$, then |A| is equal to (where |A| > 0)

Ans. 00002.00

Sol. $|\text{Adj}(A)| = |A|^{n-1} \Rightarrow 4 = |A|^2$

88. Number of solutions of the equation $\sin x + \cos x + \sin 4x + \cos 4x = 0$ in $[0, \pi]$ is/are

Ans. 00004.00

Sol. $\sin x + \cos x + \sin 4x + \cos 4x = 2 \left(\sin \left(\frac{5x}{2} \right) + \cos \left(\frac{5x}{2} \right) \right) \cos \frac{3x}{2}$
 $\Rightarrow \cos \frac{3x}{2} = 0, \sin \frac{5x}{2} + \cos \frac{5x}{2} = 0$
 $\Rightarrow x = \frac{3\pi}{10}, \frac{\pi}{3}, \frac{7\pi}{10}, \pi$

89. Let slope of a tangent at point P on the ellipse $\frac{x^2}{81} + \frac{y^2}{9} = 1$ is m such that normal at P is at a maximum distance from origin, then $|3m|$ is equal to

Ans. 00001.73

Sol. Let P be $(9 \cos \theta, 3 \sin \theta)$

$\Rightarrow$ For normal at maximum distance from origin $9 \tan \theta = 3 \cot \theta$

$$\Rightarrow \tan \theta = \pm \frac{1}{\sqrt{3}} \Rightarrow m = \pm \frac{1}{\sqrt{3}}$$

90. Minimum value of the function $f(x) = \frac{x^4 + 3x^2 + 3}{x^2 + 1}$ is equal to

Ans. 00003.00

Sol. $f(x) = x^2 + 1 + \frac{1}{x^2 + 1} + 1 \Rightarrow f(x) \geq 3$